

# **SYSCON 2025**

The 19<sup>th</sup> Annual IEEE International Systems Conference

April 7-10, 2025

**Delta Hotels Montreal  
Montreal, QC, Canada**

## **Enhancing Robot Navigation in Crowded Spaces through Systematic Strategy Selection**

Dr. Kleber Cabral, Dr. Jean-Alexis Delamer, Mr. Jefferson Silveira, Dr. Sidney N. Givigi

# Agenda

- Introduction
- What impacts robotic navigation
- How can we select a navigation strategy
- Simulation results
- Real world experiment

# Robotic Navigation



- Robotic platforms are already widely used in many applications, such as surveillance, mapping, and inspection.
- These are **complex** environments, **shared** with humans, animals, infrastructure, and other assets.
- Robots must navigate **safely** and **efficiently** to prevent collisions that could cause harm, damage, or financial loss.

# Research question

How can a robot choose the appropriate control strategy, balancing between efficiency and safety, to move in a cluttered environment?

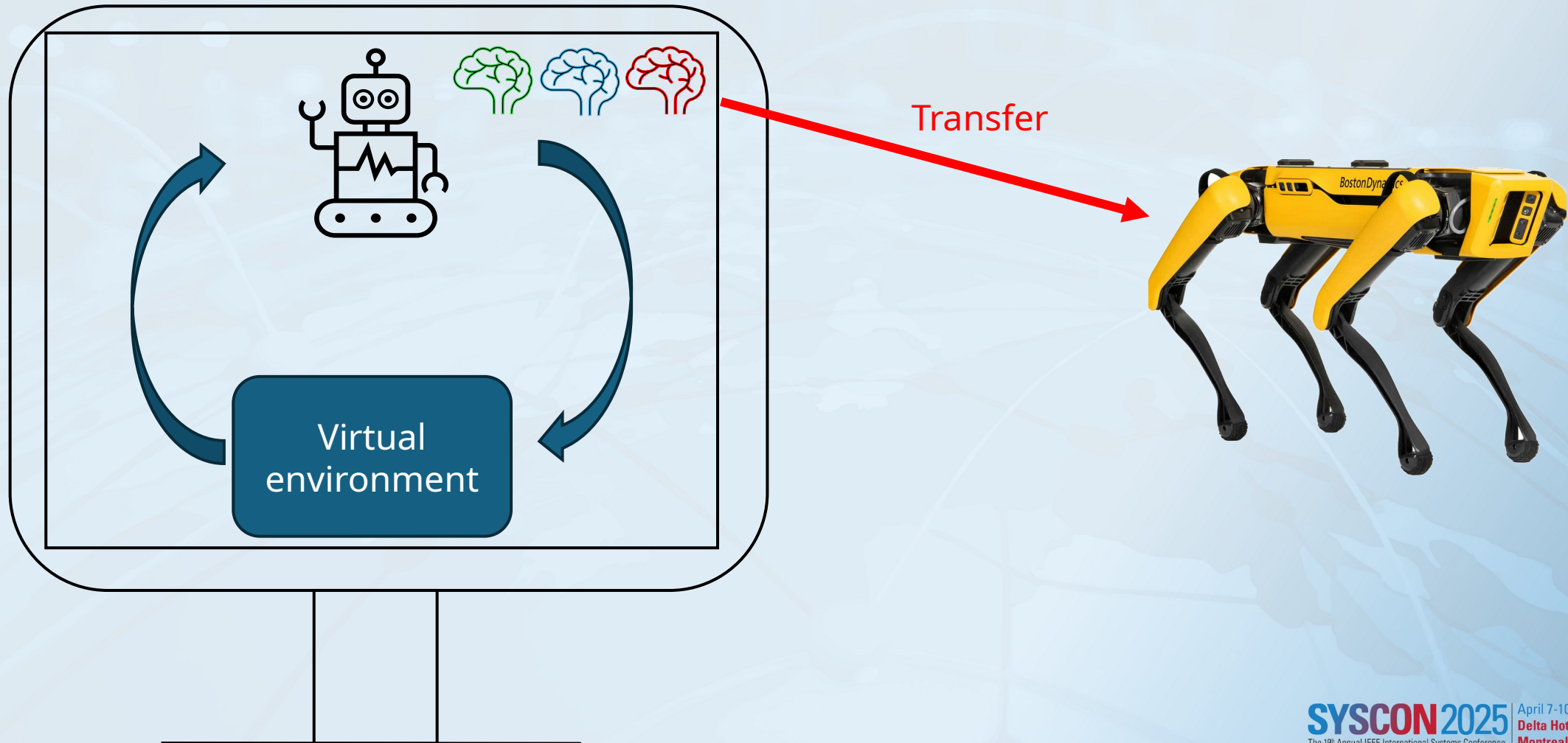


# How to assess navigation?



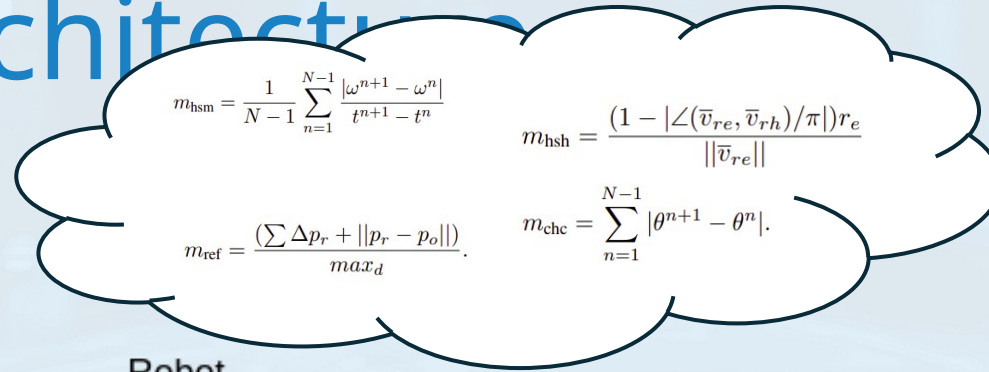
- What is safety?
- What is efficiency?
- How can they be measured?

# Solution – Dynamic Selection of AI-based Control Strategy



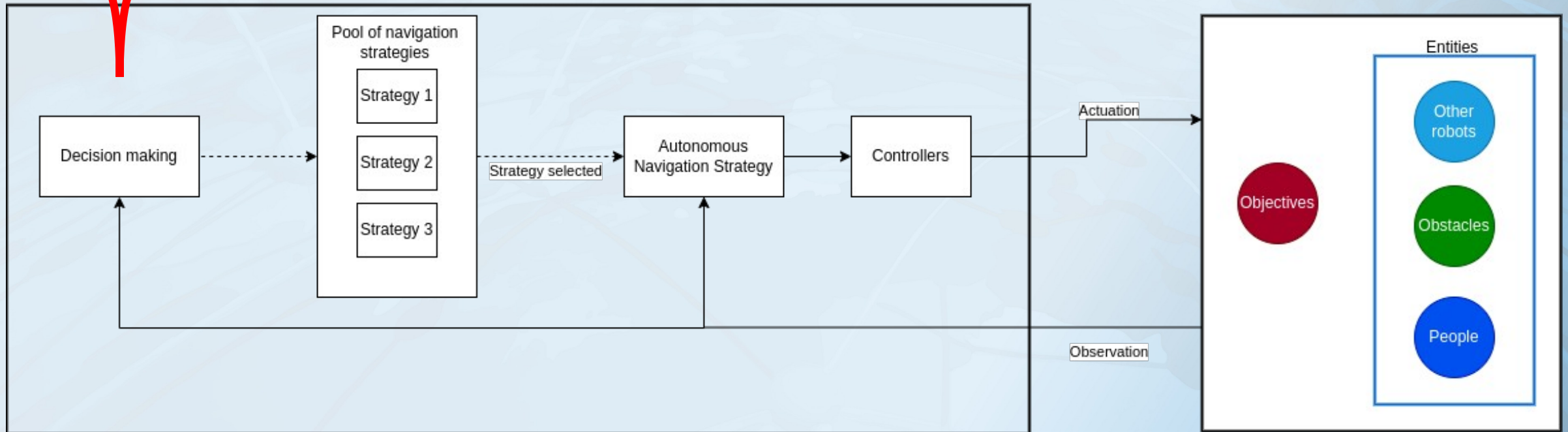
# Decision making and controller – system architecture

Safe  
Efficient  
Scary

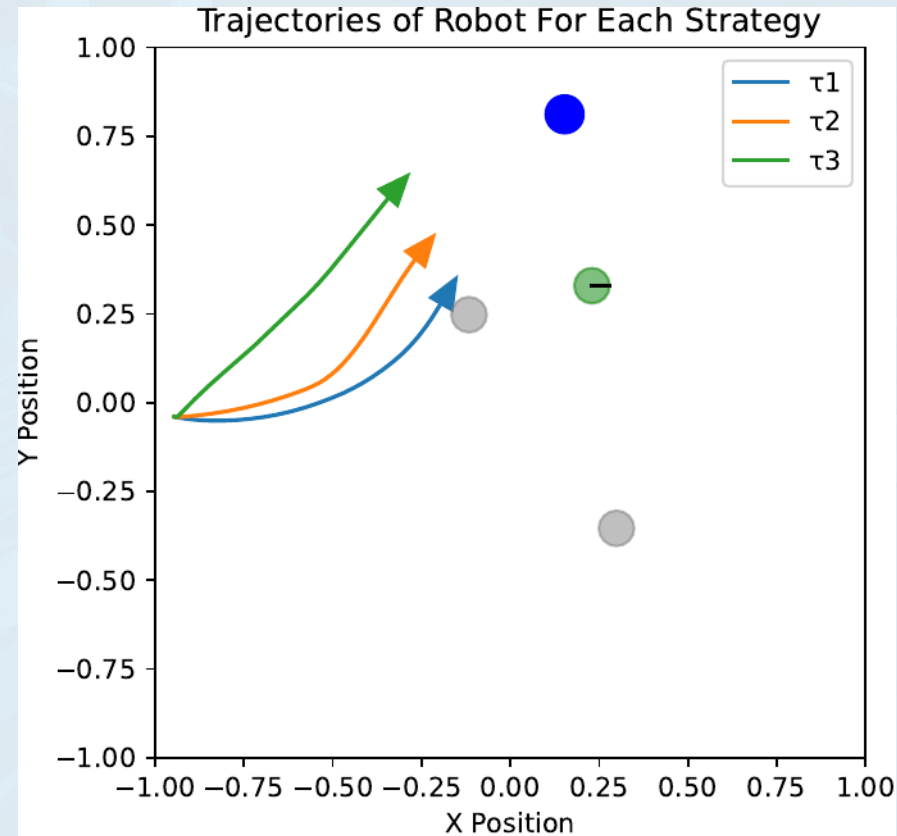


Robot

Environment



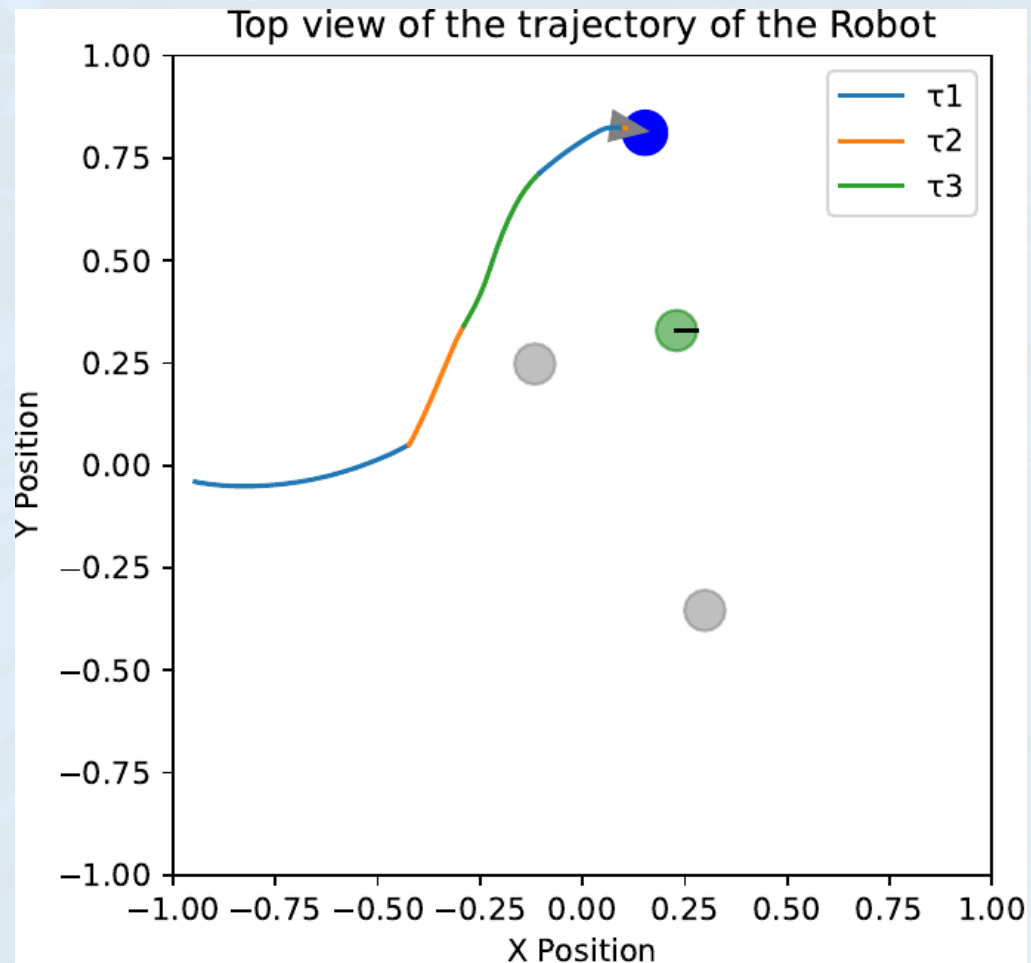
# Results



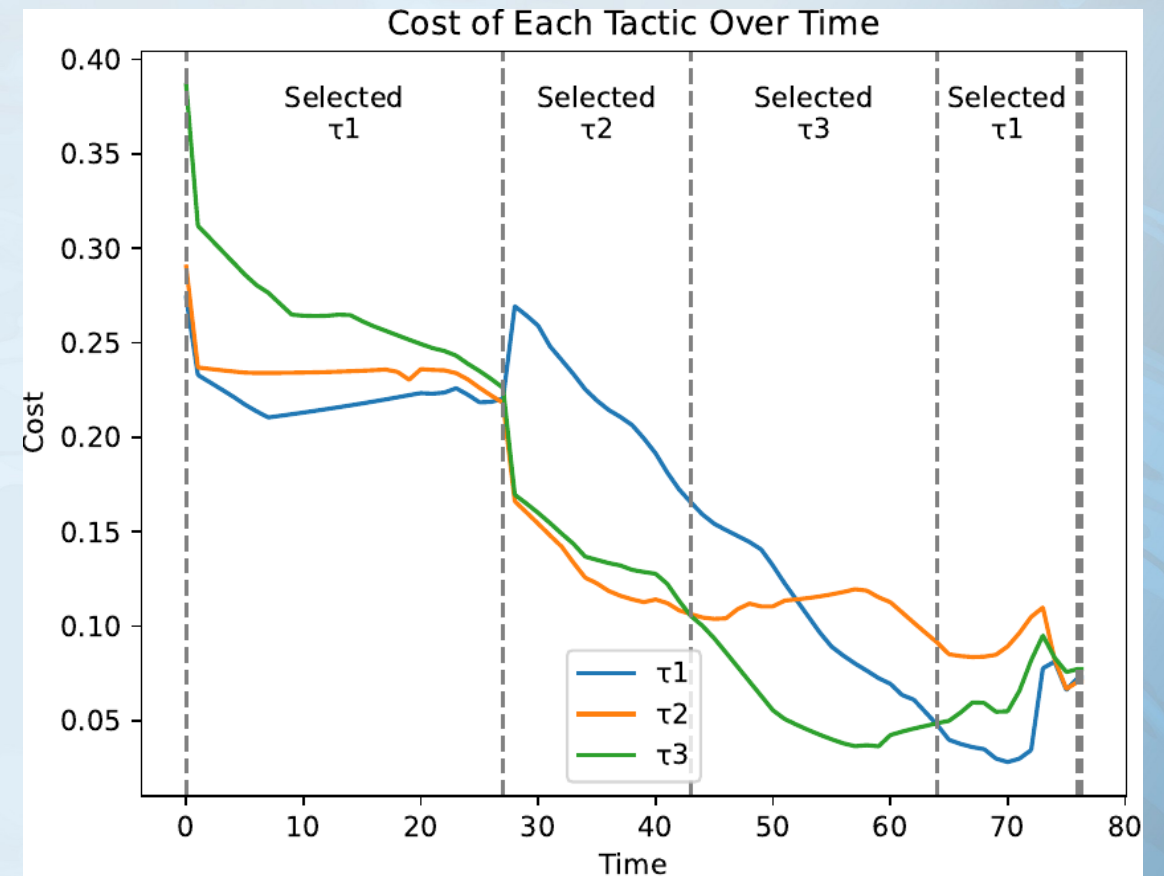
- Training – different strategies can be obtained by changing some parameters during training.

# Results

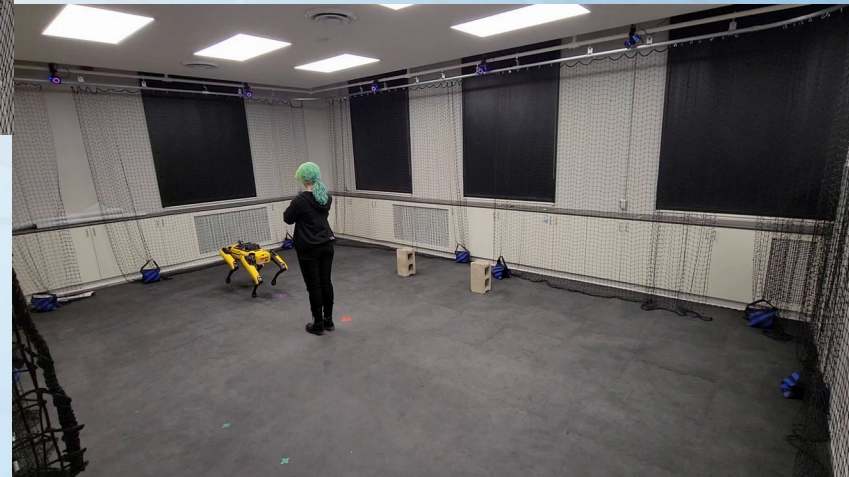
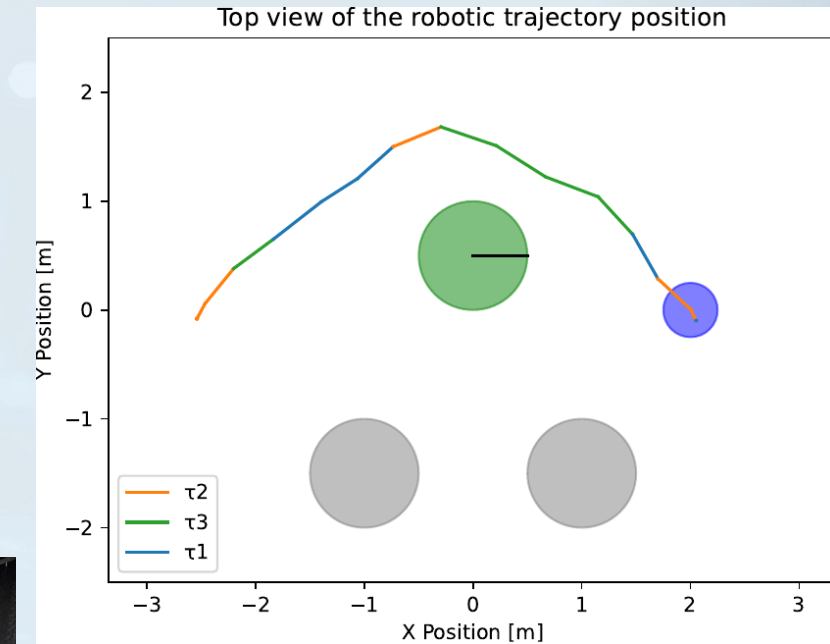
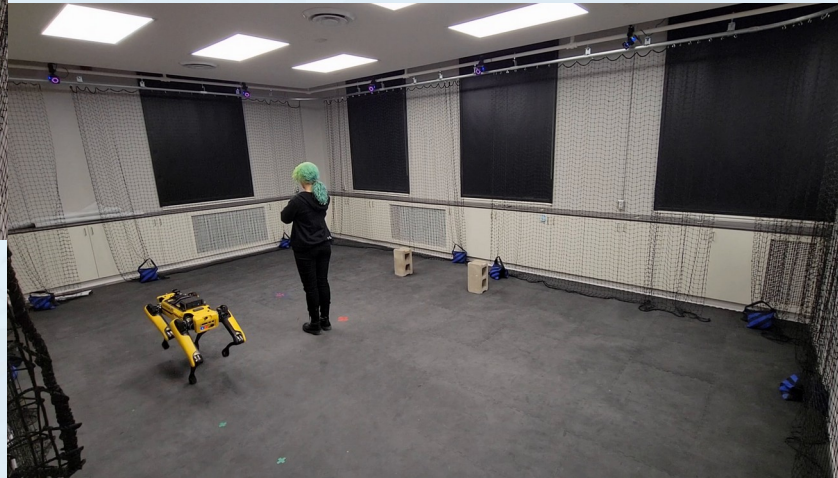
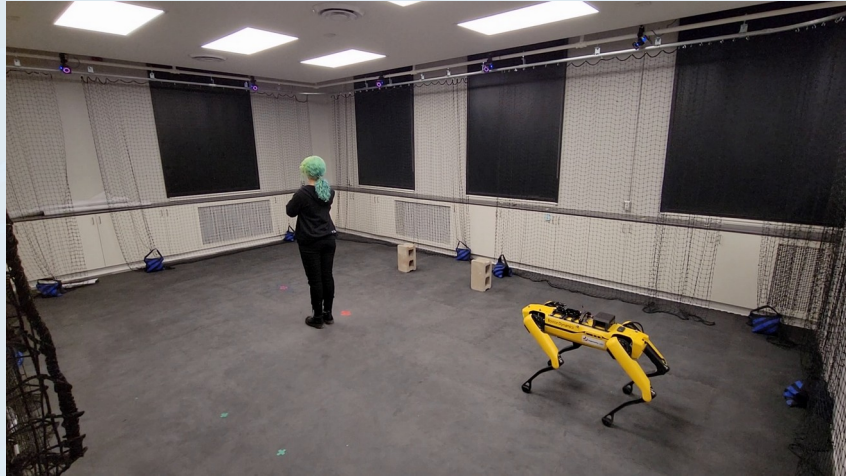
## Online selection



## Control Cost



# Experimental Results





# Thank you

# Questions?